HCI PROJECT ▪ LLM-BASED, EMOTION-ADAPTIVE MENUS

Experimental Trial Platform

Phase 1: emotion-labelled data collection

At a glance

We have built a complete, automated platform for running our trials. With a single command, it **induces an emotion**, has the participant **complete navigation tasks on a website**, records **eye tracking, GSR, heart rate and behaviour** on one shared clock, and saves everything as **model-ready, labelled data**.

3 groups, **6** emotions	**3** purpose-built websites	**72** matched tasks
≈**40 min** per participant	**1 command** per session	**10** output files per participant

1 Study design

Each group has two contrasting emotions and its own website. Every participant experiences **both** emotions of their group (a within-subjects design), so each person serves as their own baseline. This matters for a sample of 5–6 participants per group.

Group	Emotions	Contrast in the signals	Website (our own build)
G1	Stress vs. Calm	high vs. low arousal, negative vs. positive	University student portal
G2	Frustration vs. Boredom	high vs. low arousal	Online store
G3	Joy vs. Sadness	positive vs. negative valence	Discussion forum

Table 1: Groups, emotions and websites. Each group's two emotions were chosen so they are reliable to induce in a lab and clearly separable in the physiological signals.

Counterbalancing. The participant number decides the order of the two emotions and of the two task sets (A/B), following a 2×2 Latin square, so every four consecutive participants are fully balanced. In this phase every participant uses one **standard drop-down menu**: we collect clean emotion-labelled data first, and build the adaptive menus from the trained model in phase 2.

2 Session procedure

The session runs on its own. The participant follows on-screen instructions and the researcher only watches.

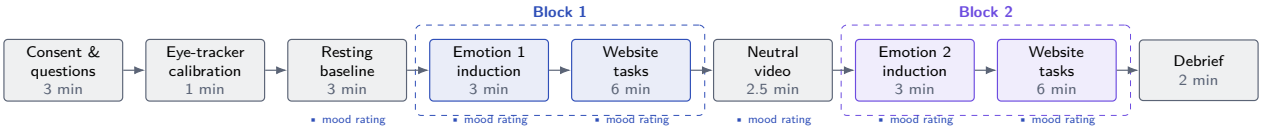


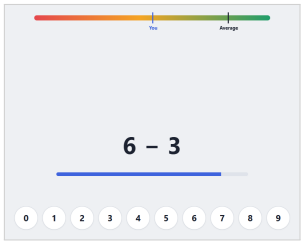
Figure 1: One session (≈40 minutes). A mood rating follows the baseline, every induction, every task block and the neutral video. Task blocks also end with a workload rating (NASA-TLX).

3 Emotion induction

Each emotion is induced with a published method *before* the task block. Where the literature supports it, a manipulation *during* the tasks keeps the state going, because film clips alone fade within minutes [13].

Emotion	Before the tasks (≈3 min)	During the tasks	Basis
Stress	Timed mental arithmetic (MIST), with a rigged "you vs. average" bar and negative feedback	Visible countdown, "compared with other participants" banner	[3, 7]
Calm	Nature video with paced breathing (6 breaths/min)	"No rush" instructions, no timer	[11, 8]
Frustration	Goal and incentive framing ("fastest participants get a bonus")	Deliberate faults: slow pages, ignored clicks, menus closing, "session expired"	[10, 2]
Boredom	Copying dull technical text	Every task uses the same repetitive template	[12, 9]
Joy	Wordless comedy clip (*Mr. Bean*, "The Exam")	Surprise "Bonus unlocked!" rewards	[4, 6]
Sadness	Sad film scene (*Taare Zameen Par*), then quiet reflection	None	[4]

Table 2: Induction methods. The film clips follow Gross and Levenson's method [4], but use culturally familiar clips. All inductions are checked with the participant's own mood ratings.



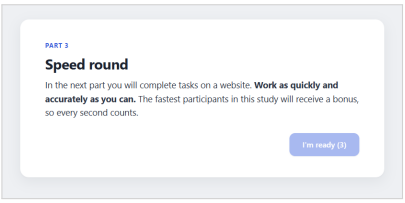
(a) Stress: MIST arithmetic



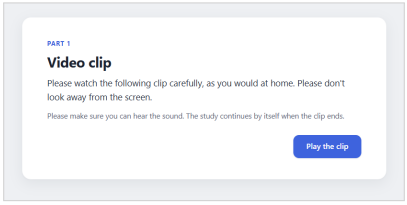
(b) Calm: nature and breathing



(c) Boredom: copying task



(d) Frustration: speed-bonus framing



(e) Joy and sadness: film clip

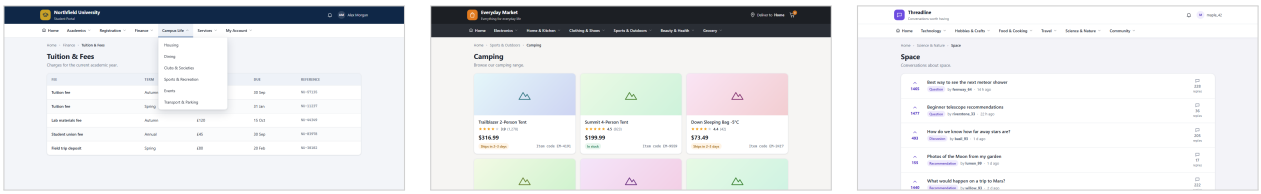


(f) Neutral video between blocks

Figure 2: Induction screens as the participant sees them.

4 Websites and tasks

The three websites are our own builds on one shared engine, so menu behaviour, timing and logging are identical across groups. Their structure is modelled on the realistic sites of the WebArena benchmark [14]: **6 sections** × **6 pages**, with the **drop-down menu as the only way to navigate**. Every page change in the data is therefore a menu interaction, which is exactly what phase 2 will adapt.



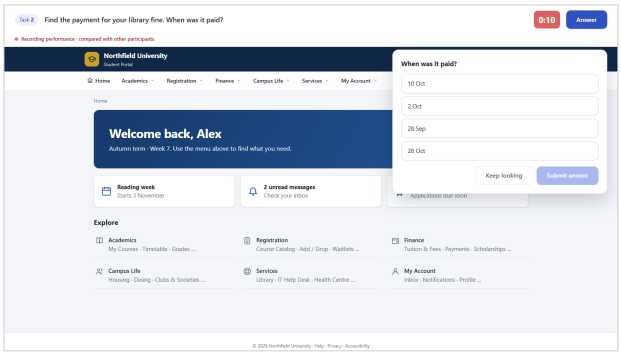
(a) G1: student portal **(b)** G2: online store **(c)** G3: discussion forum

Figure 3: The three websites (fictional brands, no internet needed).

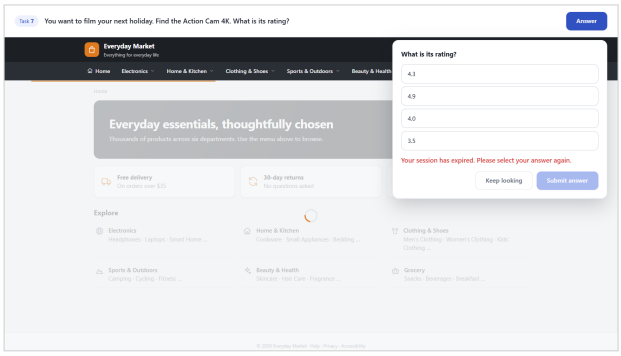
Task type	Example (G2 store)	What it measures
Look-up (8 per set)	"Find the Trailblazer 2-Person Tent. What is its price?"	Finding the right menu path from an item name
Extreme (2 per set)	"Find the cheapest vitamin or supplement."	Navigating, then scanning a page
Compare (2 per set)	"Which is cheaper: the Yoga Mat or the Vitamin C Serum?"	Navigating across several pages

Table 3: Each website has two matched task sets (A and B) of 12 tasks. Answers are multiple choice and are computed automatically from the site content.

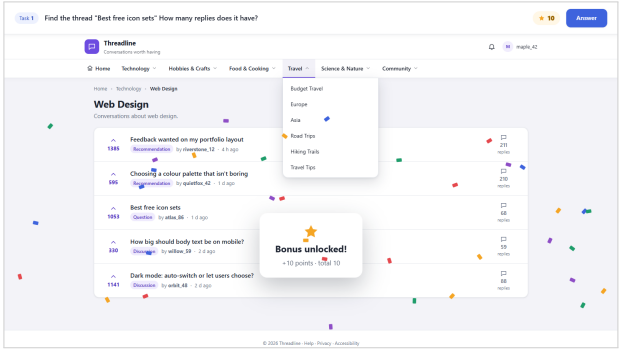
Each task block lasts exactly **6 minutes**, so every emotion yields the same amount of data per person.



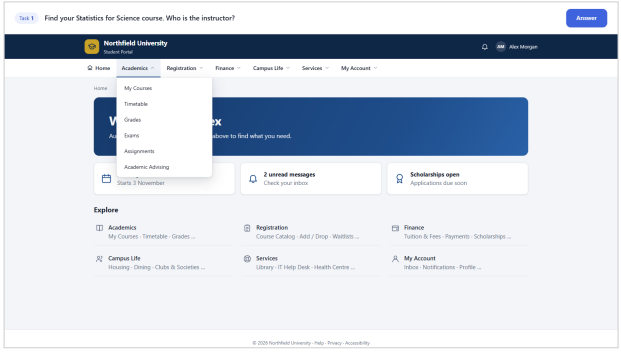
(a) Stress: countdown, recording banner, answer panel and feedback



(b) Frustration: page loading lag and a fake "session expired" error



(c) Joy: surprise bonus reward



(d) The standard drop-down menu in use

Figure 4: The task screen: the task bar sits above the live website.

5 Measures and data

Source	What is recorded
Eye tracker (Tobii Pro)	Gaze position and pupil size (60–120 Hz), mapped to menu items and page content
GSR	Skin conductance in microsiemens (supports Shimmer3, LSL devices, Arduino)
Heart rate	BPM and RR intervals (supports Polar H10, LSL, serial)
Behaviour	Every click, menu open/hover/close, page view, mouse movement (60 Hz), task outcome and time
Self-report	SAM valence/arousal/dominance [1], six emotion ratings, NASA-TLX [5]

Table 4: Measures. All streams share one master clock; the eye tracker's clock offset is re-measured every 30 s.

Each participant gets one folder, for example `data/G2/P04/`, containing the raw streams (gaze, GSR, heart rate, events, mouse, tasks, questionnaires) and `merged.csv`: a 10 Hz timeline that combines every signal with its phase, emotion and task labels, ready for model training. Table 5 shows an excerpt.

t (s)	Emotion	Task	Gaze x	Gaze y	Pupil	Looking at	GSR	HR
143.6	stress	B01	0.496	0.505	3.89	content: Mr Tom Reyes	7.46	77
143.7	stress	B01	0.494	0.499	3.90	content: Mr Tom Reyes	7.44	77
143.8	stress	B01	0.499	0.489	3.86	content: Dr Helen Moore	7.42	77

Table 5: Excerpt of `merged.csv` from a dry run with simulated sensors (selected columns).

6 Running a session

Researcher workflow

`python run.py check`	Checks sensors, videos, browser and task answers; must say "Ready."
`python run.py session --group G2 --participant P04`	Runs the full session in a fullscreen browser
`python run.py export`	Combines all participants into one dataset

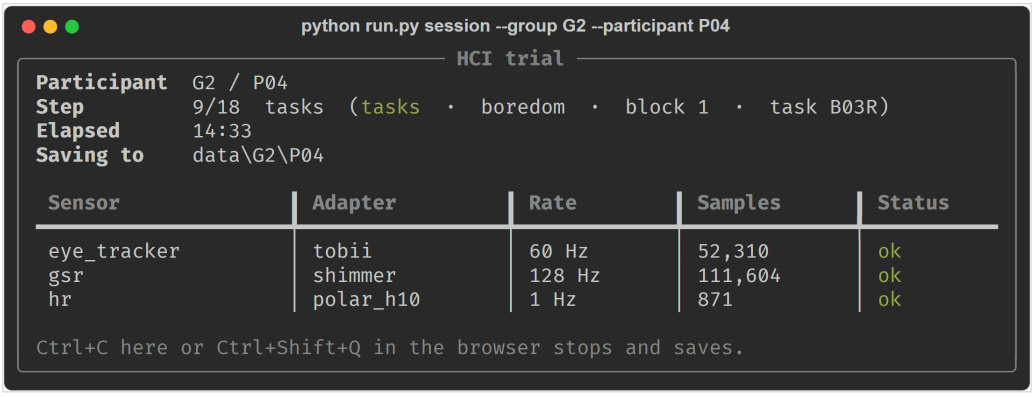


Figure 5: The researcher's live terminal view during a session: current phase, emotion, task and sensor health. The participant never sees it.

Built-in safeguards:

- A session will not start if a sensor sends no data or a video is missing.
- Every data row is saved immediately, and an interrupted session can be resumed.
- A participant ID can never be reused by accident.
- Participants never see emotion names or correct answers.

7 Ethics

- Participants sign a consent form that says, in general terms, that some activities are designed to influence how they feel.
- The on-screen debrief then explains exactly which manipulations they experienced (for example, that the website lag was deliberate) and asks again for permission to use their data.
- If the participant declines, at the start or at the debrief, **their data is deleted automatically**.
- Inductions are brief and mild, and the researcher gives a verbal wellbeing check at the end.

8 Status and next steps

Item	Status
Websites, tasks, inductions, questionnaires, debrief (all three groups)	**Done**
Videos (nature, neutral, amusement, sadness)	**Done**
Full dry runs for all groups with simulated sensors; 17 automated tests	**Passed**
Set the GSR and heart-rate device models; one dry run on the real hardware	**Next**
Ethics approval; recruit 5–6 participants per group (15–18 in total)	**Next**
Phase 2: train the emotion model and build the adaptive menus	**Later**

References

[1] M. M. Bradley and P. J. Lang. Measuring emotion: The Self-Assessment Manikin and the semantic differential. *J. Behav. Ther. Exp. Psychiatry*, 25(1):49–59, 1994.

[2] I. Ceaparu, J. Lazar, K. Bessiere, J. Robinson, and B. Shneiderman. Determining causes and severity of end-user frustration. *Int. J. Human-Computer Interaction*, 17(3):333–356, 2004.

[3] K. Dedovic et al. The Montreal Imaging Stress Task. *J. Psychiatry & Neuroscience*, 30(5):319–325, 2005.

[4] J. J. Gross and R. W. Levenson. Emotion elicitation using films. *Cognition and Emotion*, 9(1):87–108, 1995.

[5] S. G. Hart and L. E. Staveland. Development of NASA-TLX. *Advances in Psychology*, 52:139–183, 1988.

[6] A. M. Isen and P. F. Levin. Effect of feeling good on helping: Cookies and kindness. *J. Pers. Soc. Psychology*, 21(3):384–388, 1972.

[7] C. Kirschbaum, K. M. Pirke, and D. H. Hellhammer. The "Trier Social Stress Test". *Neuropsychobiology*, 28(1–2):76–81, 1993.

[8] P. M. Lehrer and R. Gevirtz. Heart rate variability biofeedback: How and why does it work? *Frontiers in Psychology*, 5:756, 2014.

[9] A. Markey, A. Chin, E. M. Vanepps, and G. Loewenstein. Identifying a reliable boredom induction. *Perceptual and Motor Skills*, 119(1):237–253, 2014.

[10] J. Scheirer, R. Fernandez, J. Klein, and R. W. Picard. Frustrating the user on purpose. *Interacting with Computers*, 14(2):93–118, 2002.

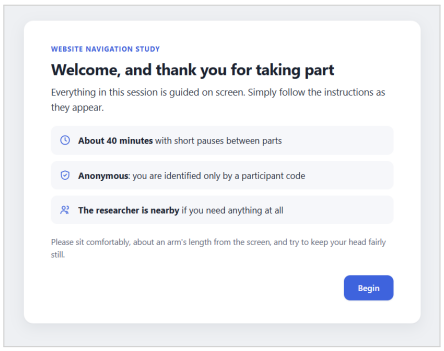
[11] R. S. Ulrich et al. Stress recovery during exposure to natural and urban environments. *J. Environmental Psychology*, 11(3):201–230, 1991.

[12] W. A. P. van Tilburg and E. R. Igou. On boredom: Lack of challenge and meaning as distinct boredom experiences. *Motivation and Emotion*, 36(2):181–194, 2012.

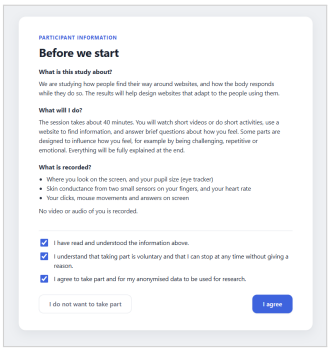
[13] R. Westermann, K. Spies, G. Stahl, and F. W. Hesse. Relative effectiveness and validity of mood induction procedures: A meta-analysis. *European J. Social Psychology*, 26(4):557–580, 1996.

[14] S. Zhou et al. WebArena: A realistic web environment for building autonomous agents. In *Proc. ICLR*, 2024.

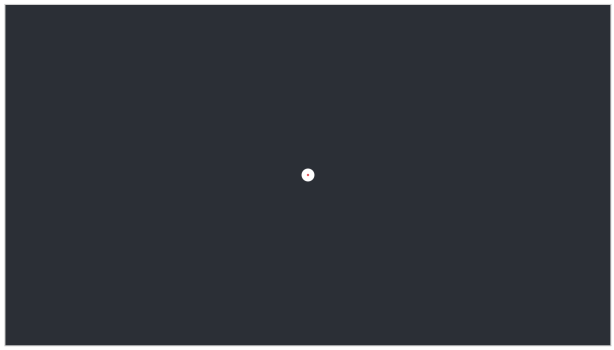
A Participant screens



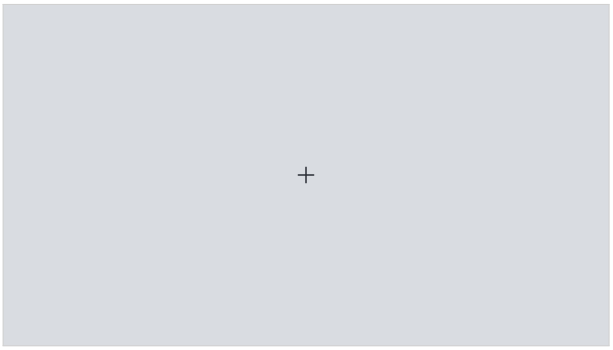
(a) Welcome



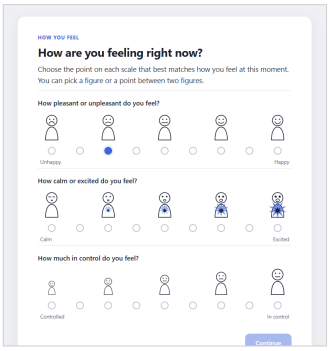
(b) On-screen consent



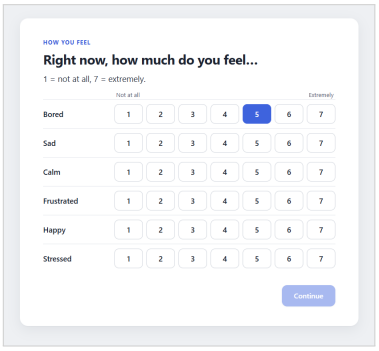
(c) Eye-tracker calibration



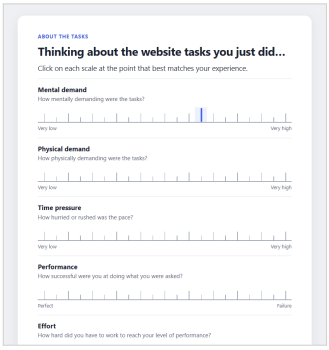
(d) Resting baseline (fixation cross)



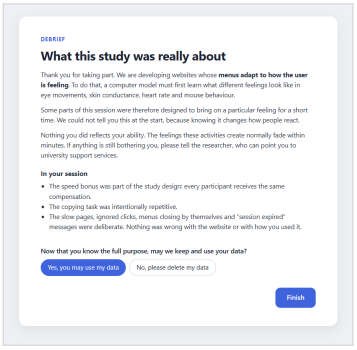
(e) Mood rating: Self-Assessment Manikin



(f) Mood rating: six emotions (order shuffled)



(g) Workload rating (NASA-TLX)



(h) Debrief, with the option to withdraw

Figure 6: The remaining screens of a session.